

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Programming in C (New)

Semester: Fall

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain the key differences between machine-level language, assembly language, and high-level language. 8
b) Define algorithm and flowchart. Draw a flowchart that takes a student's marks and classifies the grade as: 7
 - i. Marks $\geq 90 \rightarrow A$
 - ii. Marks: 80 to 89 $\rightarrow B$
 - iii. Marks: 70 to 79 $\rightarrow C$
 - iv. Marks $< 70 \rightarrow \text{Fail}$
2. a) What are keywords and identifiers? List out the naming rules for identifiers. 7
b) Why do we need different data types in programming? Describe the four basic data types along with their size and range. 8
3. a) Explain pre-test and post-test loops with suitable program examples and also draw a flowchart. 8
b) Compare break and continue statements. Write a program to find the sum of all numbers greater than 100 and less than 300 that are divisible by 5. 7
4. a) Write a program to find the sum of two 3 x 3 matrices. 7
b) How are strings implemented in C? Write a program to check a string is palindrome or not. 8
5. a) Write a function in C that calculates the factorial of a number using recursion. Explain how recursion works in your code. 7

OR

Explain static, local and global variables in c with examples.

- b) Write about different components of a function with syntax. Write a program to print Fibonacci series up to n terms using concept of recursive function. 8
6. a) Explain Pointer, NULL Pointer and Generic Pointer. Write a program to read an array of 5 elements and find its largest element using concept of pointer. 7

OR

Explain the functions used in dynamic memory allocation (DMA). Write a program to find sum and average of n numbers using concept of DMA.

- b) Create a structure called students which contains student's name, id, age and marks. Read the marks scored by 20 students and display the records whose mark is greater than 60. 8
7. Write short notes on: (**Any two**) 2×5
- a) Call by reference
- b) File opening functions
- c) File handling modes